The baseline energy trajectory is the blue energy line in Fig.1 where charging starts immediately at the rated power upon plug-in until the expected energy level is reached. The actual charging trajectory is the red line in Fig.1. The area within the upward and downward energy adjustment ranges of actual charging trajectory represents the flexible reserves in accumulated energy consumption.

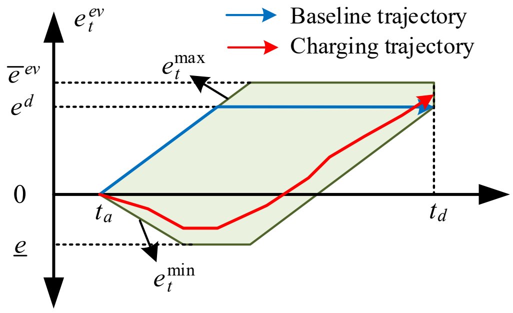

Fig.1. The flexibility and baseline of EV accumulated energy consumption.

Meanwhile, the flexibility cost of one EV is calculated by:

$$C^{ev} = \sum_t \left(c^{ev+} \left[e_t - e_t^{base} \right]^+ + c^{ev-} \left[e_t^{base} - e_t \right]^+ \right) \tag{1.1}$$